Please refer to pages **8–21** of the recommended text book.

Unit 1.3

What is function?

Definition of a function. (please refer to page 8 of the recommended text book.

Representing functions. (please see page 10–12 of the text book)

verbally: by description in words

numerically: by a table of values

visually: by a graph

algebraically: by an explicit formula

Finding Domain and range of a function. Consider the following functions

(1) $f(x) = \sqrt{1-x}$

(2) $g(t) = \frac{1}{t^2}$

(3) $A(r) = \pi r^2$

(4) $h(u) = 5^u$

Piecewise defined functions (see page 14). Consider the following functions and discuss their domain and range

(1) $f(x) = \begin{cases} 4 & \text{if } \ x < -2 \\ -x^2 + 1 & \text{if } \ -2 \leq x < 1 \\ x + 4 & \text{if } \ x > 1 \end{cases}$

(2) $h(x) = x + |x + 1|$

Sketching graph of $y = |f(x)|$ from the graph of $y = f(x)$.

Consider the following examples

(1) $y = |x + 1|$

(2) $y = |x^2 + 1|$

(3) $y = |-2^x|$

Tips to Identify rules which define functions (see page 13).

The Vertical line test:

Determine whether or not $y = f(x)$ defines a function

- $x^2 + y^2 = 4$

- $x = y^2$

Even and Odd functions or Symmetric functions (See page 16).

Even functions: Symmetric with respect to the $y-$axis

Odd functions: Symmetric with respect to the origin

Increasing and decreasing functions.

Increasing functions:

Decreasing functions: